THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The decimal value of 11011_2 is: 1

(a) 33 **(b)** 29 **(c)** 27 **(d)** 25

2. The angles of a quadrilateral are x°, $(x+20)^\circ$, $(x+40)^\circ$ and $(x+60)^\circ$. The value of x is: 1

(a) 80 **(b)** 40 **(c)** 60 **(d)** 70

3. The digit that replaces * in 6*81 so that the number is divisible by 11 is: 1

(a) 3 **(b)** 5 **(c)** 2 **(d)** 8

4. $(4x+1)(3x+2)$ equals: 1

(a) $12x^2+11x-2$ **(b)** $12x^2-11x+2$ **(c)** $12x^2+11x+2$ **(d)** $12x^2+2$

5. Two numbers are in the ratio 5 : 8 and their sum is 78. The larger number is: 1

(a) 48 **(b)** 30 **(c)** 39 **(d)** 54

6. An article is sold for ₹ 2400 at a profit of 20%. Its cost price (in ₹) is: 1

(a) 2000 **(b)** 1920 **(c)** 1800 **(d)** 2200

7. A rectangle has a diagonal of 39 cm and one side 15 cm. Its perimeter (in cm) is: 1

(a) 104 **(b)** 51 **(c)** 108 **(d)** 102

8. If A:B = 2:7 and B:C = 14:5, then A:C is: 1

(a) 2:3 **(b)** 10:98 **(c)** 2:5 **(d)** 4:5

9. A prism has 24 edges. The number of its faces is: 1

(a) 11 **(b)** 10 **(c)** 8 **(d)** 9

10. The mean of 20, 39, 32, 34, 52, 27 is 34. If 32 is replaced by 44, the new mean is: 1

(a) 46 **(b)** 34 **(c)** 41.33 **(d)** 36

11. $(x+3)(x+4)$ equals: 1

(a) $2x+7$ **(b)** $x^2+12x+7$ **(c)** x^2+7 **(d)** $x^2+7x+12$

12. A parallelogram has base 20 cm and height 12 cm. If the base is changed to 16 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 15 **(b)** 18 **(c)** 224 **(d)** 14

13. The ones digit of 238^2 is: 1

(a) 8 **(b)** 6 **(c)** 2 **(d)** 4

14. $\frac{5^7 \times 5^3}{5^6}$ equals: 1

(a) 5^{10} **(b)** 5^4 **(c)** 5^{16} **(d)** 5^{15}

15. The binary form of 54 is: 1

(a) 011011 **(b)** 110101 **(c)** 110111 **(d)** 110110

16. The adjacent sides of a parallelogram are in the ratio 2:3 and its perimeter is 60 cm. The longer side is: 1

(a) 30 cm **(b)** 18 cm **(c)** 36 cm **(d)** 12 cm

17. The smallest digit that replaces * in 71*4 so that the number is divisible by 6 is: 1

(a) 1 **(b)** 0 **(c)** 5 **(d)** 2

18. $(3x - y)^2$ equals: 1

(a) $9x^2 - 6xy + y^2$ **(b)** $9x^2 - y^2$ **(c)** $9x^2 - 3xy + y^2$ **(d)** $9x^2 + 6xy + y^2$

19. Two numbers are in the ratio 2 : 7. If 3 is added to each, the numbers become 15 and 45. The original smaller number is: 1

(a) 12 **(b)** 9 **(c)** 15 **(d)** 24

20. A watch is sold for ₹ 2800 at a loss of 20%. Its cost price (in ₹) is: 1

(a) 3500 **(b)** 3150 **(c)** 3850 **(d)** 3360

21. The side of a rhombus is 39 cm and one diagonal is 30 cm. The other diagonal (in cm) is: 1

(a) 48 **(b)** 74 **(c)** 36 **(d)** 72

22. 9 workers working 6 hours a day finish a job in 20 days. 15 workers working 4 hours a day will finish it in (days): 1

(a) 12 **(b)** 18 **(c)** 180 **(d)** 20

23. A pyramid has 9 vertices. The number of its edges is: 1

(a) 16 **(b)** 8 **(c)** 10 **(d)** 18

24. The median of 49, 15, 16, 12, 98, 66, 92, 32 is: 1

(a) 40.5 **(b)** 33 **(c)** 98 **(d)** 47.5

25. If $a = 12$ and $b = 3$, the value of $4a - 6b + 10$ is: 1

(a) 80 **(b)** 36 **(c)** 40 **(d)** 44

26. A rhombus has side 26 cm and one diagonal 20 cm. Its area is: 1

(a) 480 cm² **(b)** 104 cm² **(c)** 240 cm² **(d)** 520 cm²

27. $\sqrt{2.25}$ is equal to: 1

(a) 1.6 **(b)** 0.15 **(c)** 15.0 **(d)** 1.5

28. The value of $\left(\frac{3}{5}\right)^{-2}$ is: 1

(a) $\frac{25}{9}$ **(b)** $\frac{10}{3}$ **(c)** $\frac{9}{25}$ **(d)** $-\frac{25}{9}$

29. LXXVIII + CXXXIV equals: 1

(a) CCXII **(b)** CCXXII **(c)** CCII **(d)** CCXIII

30. The diagonals of a parallelogram ABCD meet at O. If AO $= (3x + 1)$ cm and OC $= (5x - 5)$ cm, then AC is: 1

(a) 22 cm **(b)** 10 cm **(c)** 20 cm **(d)** 3 cm

31. One step of the Kaprekar routine applied to 6230 gives: 1

(a) 6102 **(b)** 6084 **(c)** 5985 **(d)** 6183

32. The value of $78^2 - 22^2$ is: 1

(a) 5600 **(b)** 3136 **(c)** 10000 **(d)** 6568

33. A mixture of 84 L has milk and water in the ratio 5 : 2. The quantity of water is: 1

(a) 29 L **(b)** 60 L **(c)** 19 L **(d)** 24 L

34. After a discount of 20%, a pair of shoes is sold for ₹ 2240. The marked price (in ₹) is: 1

(a) 2688 **(b)** 3080 **(c)** 2520 **(d)** 2800

35. Two vertical poles of heights 35 m and 20 m stand 36 m apart on level ground. The distance between their tops (in m) is: 1

(a) 37 **(b)** 40 **(c)** 39 **(d)** 51

36. The ratio of boys to girls in a school is 5:8. If there are 780 students, the number of girls is: 1

(a) 480 **(b)** 390 **(c)** 300 **(d)** 767

37. All 12 faces of a polyhedron are triangles. The number of its edges is: 1

(a) 24 **(b)** 12 **(c)** 18 **(d)** 36

38. The mean of 5 numbers is 30. The number to be added so that the mean becomes 25 is: 1

(a) -5 **(b)** 0 **(c)** 25 **(d)** 5

39. A rectangle has length $(4x + 1)$ cm and breadth $(3x + 6)$ cm. Its perimeter (cm) is: 1

(a) $14x + 14$ **(b)** $12x^2 + 6$ **(c)** $7x + 7$ **(d)** $14x + 7$

40. A path 3 m wide runs around the outside of a 30 m by 18 m rectangular garden. The area of the path is: 1

(a) 36 m² **(b)** 324 m² **(c)** 288 m² **(d)** 864 m²

41. $\sqrt[3]{0.125}$ is equal to: 1

(a) 5.0 **(b)** 0.05 **(c)** 0.5 **(d)** 0.6

42. If $5^x \times 5^2 = 5^9$, then x is: 1

(a) 11 **(b)** 9 **(c)** 7 **(d)** 18

43. The largest 3-digit number in base 3, written in decimal, is: 1

(a) 27 **(b)** 26 **(c)** 25 **(d)** 8

44. In a rhombus ABCD, $\angle A = (3x + 15)^\circ$ and $\angle B = (6x + 39)^\circ$. The value of x is: 1

(a) 14 **(b)** 18 **(c)** 13 **(d)** 15

45. The smallest number divisible by 6, 8 and 12 is: 1

(a) 24 **(b)** 48 **(c)** 12 **(d)** 30

46. The value of $(x + 4)^2$ when $x = 6$ is: 1

(a) 52 **(b)** 76 **(c)** 108 **(d)** 100

47. 8 pipes fill a tank in 9 hours. 12 such pipes will fill it in: 1

(a) 5 hours **(b)** 9 hours **(c)** 6 hours **(d)** 72 hours

48. The simple interest on a sum at 8% p.a. for 2 years is ₹ 720. The sum (in ₹) is: 1

(a) 4950 **(b)** 4050 **(c)** 9000 **(d)** 4500

49. The area of a right triangle with hypotenuse 39 cm and one leg 15 cm (in cm^2) is: 1

(a) 540 **(b)** 285 **(c)** 292 **(d)** 270

50. A sum of ₹ 1080 buys 30 kg of rice at ₹ 36 per kg. When the price falls to ₹ 45 per kg, the same money buys (kg): 1

(a) 60 **(b)** 24 **(c)** 30 **(d)** 22

END OF QUESTION PAPER **Total: 50 Marks**

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